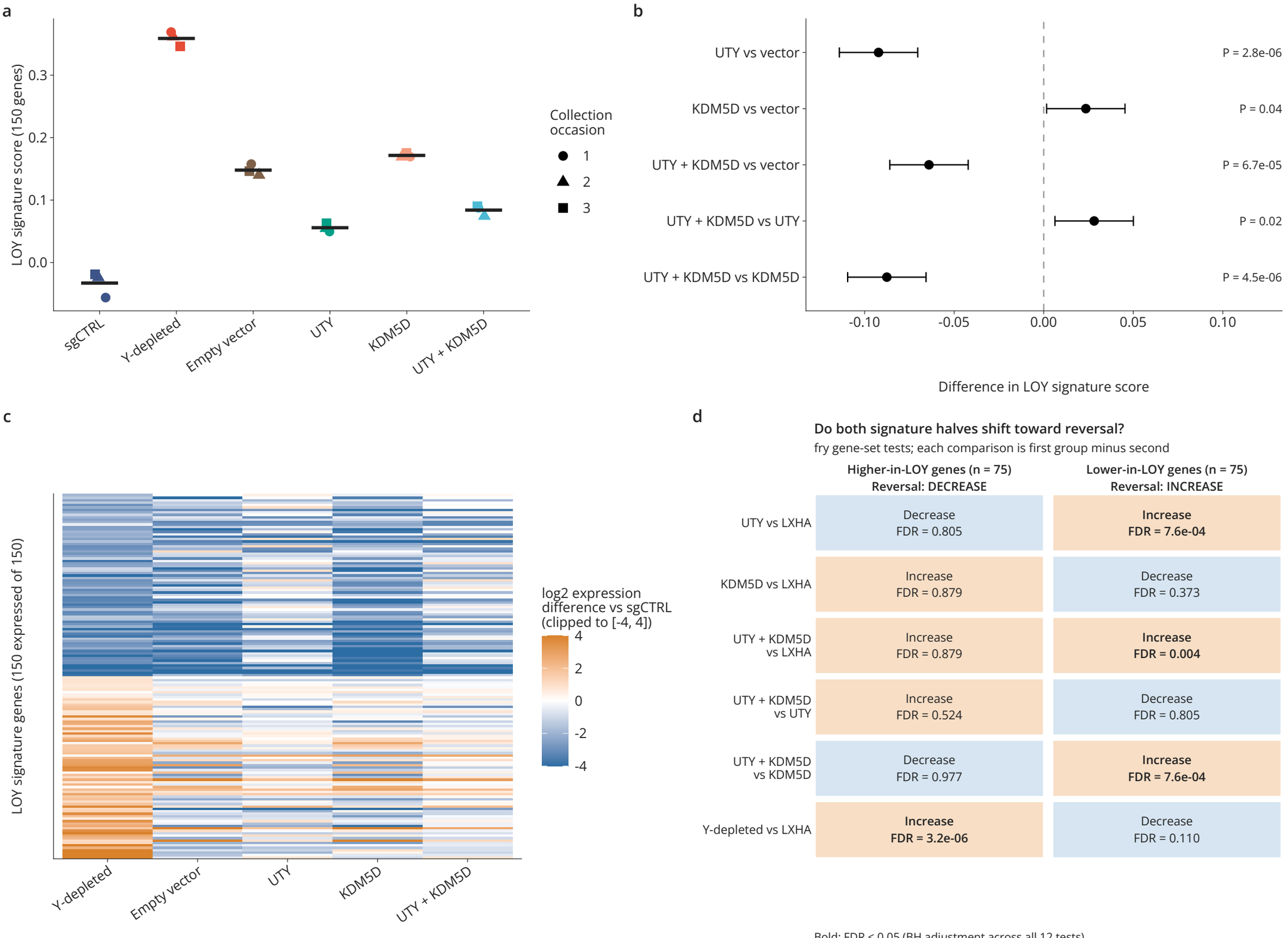

a
LOY signature score (150 genes)
0.3
0.2
0.1
0.0
sgCTRL
Y-depleted
Empty vector
UTY
KDM5D
UTY + KDM5D
Collection occasion
1
2
3
b
UTY vs vector
KDM5D vs vector
UTY + KDM5D vs vector
UTY + KDM5D vs UTY
UTY + KDM5D vs KDM5D
P = 2.8e-06
P = 0.04
P = 6.7e-05
P = 0.02
P = 4.5e-06
-0.10
-0.05
0.00
0.05
0.10
Difference in LOY signature score
c
LOY signature genes (150 expressed of 150)
Y-depleted
Empty vector
UTY
KDM5D
UTY + KDM5D
log2 expression difference vs sgCTRL (clipped to [-4, 4])
4
2
0
-2
-4
d
Do both signature halves shift toward reversal?
fry gene-set tests; each comparison is first group minus second
Higher-in-LOY genes (n = 75) Reversal: DECREASE
Lower-in-LOY genes (n = 75) Reversal: INCREASE
UTY vs LXHA
Decrease FDR = 0.805
Increase FDR = 7.6e-04
KDM5D vs LXHA
Increase FDR = 0.879
Decrease FDR = 0.373
UTY + KDM5D vs LXHA
Increase FDR = 0.879
Increase FDR = 0.004
UTY + KDM5D vs UTY
Increase FDR = 0.524
Decrease FDR = 0.805
UTY + KDM5D vs KDM5D
Decrease FDR = 0.977
Increase FDR = 7.6e-04
Y-depleted vs LXHA
Increase FDR = 3.2e-06
Decrease FDR = 0.110
Bold: FDR < 0.05 (BH adjustment across all 12 tests).
Direction and significance are distinct; color indicates direction only.